

VENKATESWARLU KOMMINENI

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EDUCATION

SRM University

Bachelor of Technology in Computer Science & Engineering

Sonipat, Delhi NCR

August 2023 – July 2027

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Object-Oriented Programming, Machine Learning

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS, Three.js

Backend & Databases: Node.js, Express.js, MongoDB, MySQL, RESTful APIs, JWT

AI & Machine Learning: TensorFlow, Keras, OpenCV, MediaPipe, Scikit-learn, NumPy, Pandas

Developer Tools: Git, GitHub, Docker, Postman, Cloudinary, Linux

EXPERIENCE

Sysslan IT Solutions

Full-Stack Web Development Intern

Remote

March 2026 – April 2026

- Architected modular React/Tailwind UI components with memoized hooks, cutting redundant DOM re-renders by 28% and boosting Lighthouse performance from 72 to 91.
- Constructed 12+ secure RESTful API endpoints in Node.js/Express for multi-step form data, integrating Joi schema validation and centralized error-handling middleware.
- Containerized backend services via Docker and configured Postman API test collections, catching 15+ edge-case integration defects prior to staging deployment.

PROJECTS

Hospital Medical System (Sushuruth) | JavaScript, Node.js, Express.js, MongoDB

[\[Repo\]](#)

- Architected an end-to-end hospital management platform handling patient admissions, digital medical records (EMR), and scheduled doctor consultations.
- Implemented role-based access control (RBAC) across admin, doctor, and patient dashboards to ensure strict medical record privacy and authorization.
- Designed optimized MongoDB aggregation schemas to query patient histories and pharmacy stock statuses with low query turnaround time.

Vahan Bazar (Two-Wheeler Marketplace) | React, Node.js, Express.js, MongoDB

[\[Repo\]](#)

- Built a full-stack e-commerce marketplace supporting multi-criteria vehicle searches, structured seller workflows, and inventories across 1,200+ indexed listings.
- Designed MongoDB aggregation pipelines and compound indexes on brand, price, and location, slashing search query latency from 850ms down to 540ms (~36% speedup).
- Integrated Cloudinary multipart image transformations and client-side cursor pagination, reducing initial page network payload weights from 3.2 MB down to 1.7 MB (47% cut).

Brain Tumor Detection AI | Python, TensorFlow, Keras, OpenCV, Scikit-learn

[\[Repo\]](#)

- Trained a deep Convolutional Neural Network (CNN) across 3,000+ axial brain MRI images from the Kaggle Br35H dataset to classify tumorous vs. non-tumorous tissue.
- Constructed an end-to-end data pipeline incorporating OpenCV contrast enhancement, scaling, and affine augmentations to prevent overfitting on clinical training subsets.
- Attained 94.2% test accuracy, 0.93 F1-score, and 95.1% sensitivity, verified via 5-fold cross-validation and confusion matrix diagnostics.

Real-Time Indian Sign Language & Emotion Recognition | Python, OpenCV, MediaPipe, Scikit-learn

[\[Repo\]](#)

- Constructed an inference pipeline translating continuous Indian Sign Language gestures into English text at 28+ FPS with under 35ms end-to-end frame latency.
- Extracted 21 coordinate landmarks per hand in 3D Euclidean space via MediaPipe Hands, feeding a normalized classifier achieving 96.4% gesture accuracy.
- Engineered a dynamic frame capture loop and adaptive bounding box stabilizer in OpenCV to eliminate tracking jitter under variable lighting.

CERTIFICATIONS